The selected head bound

$|Z_{\mathrm{loc}}| \le \frac{1}{2}$ is a census, not an unsigned Chebyshev theorem

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Abstract

The selected-sequence census of r428 left $|Z_{\mathrm{loc}}(W_k)| \le \frac{1}{2}$ as the sharpest remaining sublemma. The identity $Z_{\mathrm{loc}} = t_{\mathrm{loc}} + \mathrm{chain}$ is exact (Lean `canonical_split`). The unsigned triangle on the geometric edge, a Chebyshev majorant, and a mild left–right Abel sum all fail to produce an explicit k_0, even on the selected windows: the triangle grows (2.24 to 19.24, two-point slope +0.72). The bound remains a sealed finite census ($k = 3$ tight). No RH claim.

Status. Lemma 1 is **reduziert** (identity SATZ; unsigned/Chebyshev/Abel route REFUTED; bound a finite census; no k_0). Proposition 1 is proved. Proposition 2 is a named refutation. Ausgang `FORMULA_SATZ / TRIANGLE_REFUTED / BOUND_CENSUS / NO_K0`. No L^* claim. No $R^\dagger$ claim. No RH claim.

1 The lemma

Lemma 1 (selected head bound)**.** *$|Z_{\mathrm{loc}}(W_k)| \le \frac{1}{2}$ for every window of the selected sequence $a_k = 2^k$, $k \ge 3$.*

Verdict: REDUZIERT as posted. The identity below is a theorem. The hoped unconditional estimate is not. Measured on $k = 3, 4, 5, 6, 7, 9$ (builder pin at $k = 8$, $N_w = 5690$).

2 What Z_{loc} is

On a canonical window the terminal drive is $Z = t_{N-2} + \mathrm{chain}$ with $\mathrm{chain} = \alpha_{N-2} r_{N-2} + \beta_{N-2} r_{N-3}$. The smooth-border reconstruction $t_{N-2} = \sum_j w_j\, x_j\, v^{(2)}(x_j)\,\mathrm{fac}$ splits by the frozen hull-edge mask $E = \{x \le \ell + F(\mathfrak{h} - \ell)\} \cup \{x \ge \mathfrak{h} - F(\mathfrak{h} - \ell)\}$ at $F = \frac{1}{5}$:

$$t_{\mathrm{loc}} = \sum_{x_j \in E} w_j\, x_j\, v^{(2)}(x_j)\,\mathrm{fac},$$
$$Z_{\mathrm{loc}} = t_{\mathrm{loc}} + \mathrm{chain},$$
$$Z = Z_{\mathrm{loc}} + t_{\mathrm{bulk}}.$$

The hull of every selected window in this round is $[-1, 1]$ (up to a $2 \cdot 10^{-4}$ right endpoint). E is a *geometric* tail, not the set of small prime powers.

Proposition 1 (head identity)**.** *$Z_{\mathrm{loc}} = t_{\mathrm{loc}} + \mathrm{chain}$ and $Z = Z_{\mathrm{loc}} + t_{\mathrm{bulk}}$ hold identically (reconstruction 10^{-14}; Lean `canonical_split`). On the three live pins $k = 3, 4, 5$ the formula reproduces 0.486846, 0.157211, 0.227187. At $k = 4$, $t_{\mathrm{loc}} = -0.03720$ and $\mathrm{chain} = -0.12001$.*

n_{edge} on selected is $85, 217, 113, 429, 989, 1691$: this is not a finite von Mangoldt head.

3 The unsigned estimate

Proposition 2 (triangle grows)**.** *The unsigned edge triangle* tri $= \sum_{x_j \in E} |ct_j|$ *on selected occupies* [2.240, 19.245]*, every value* $> \frac{1}{2}$*. The two-point log-log slope against* N_w *(*$k = 3$ *vs* $k = 9$*) is* $+0.719$*. A Chebyshev majorant* $\psi(x) \le 1.04x$ *cannot close: the support of* E *is a positive fraction of a hull of width* 2*, not an interval* $[1, X_{\text{head}}]$*. Left–right pairing fails as a SATZ (*$\text{pair}_c = 1$ *on* $k = 5, 7, 9$*). Abel partial sums of the edge weights have* $|A|_{\max}/\|w\|_1 = 1$ *on* $k = 3, 4, 6$*; the phase* $\varphi = xv^{(2)}\text{fac}$ *has variation* 10^3*–*10^6*. There is no explicit* k_0*.*

The signed $|Z_{\text{loc}}|$ itself falls versus N_w on this list (census, not SATZ) because of cancellation of factor 0.013–0.145. Selected is *not* nicer for the unsigned bound than the old 42-rung ladder (r386 slope +0.25, triangle up to 14.5); here the triangle grows faster.

4 Conditional terminal balance

If one *assumes* the census limbs below, COMPOSE$^-$ (r378, Lean SATZ) gives $q_N < 1$ on the live selected rungs:

1. $|Z_{\text{loc}}| \le \frac{1}{2}$ on selected $k \in \{3, 4, 5, 6, 7, 9\}$ — CENSUS, tight at $k = 3$ (air 0.013), scramble-unstable (0.524).
2. $R = S_F/D \le \frac{3}{2}$ — CENSUS (max 1.396 at $k = 4$).
3. $L_1 \le \frac{2}{3}$ — CENSUS. The hoped $L_1 \le \frac{3}{5}$ FAILS at $k = 9$ (0.628).
4. $M_3 \le \varphi(m)$ — not re-opened (r428: under the r306 floor).
5. $k = 8$ — builder pin, not a live four-coordinate row.
6. $k \to \infty$ — OPEN ($k = 10$ border-fails, r421).

No limb except COMPOSE$^-$ and the head identity is a SATZ. This is the honest terminal balance after r429: the mincut is unchanged (base 4 / refined 5).

5 Kills

World	$\|Z_{\text{loc}}\|$ / triangle	Role
scramble seed 1 at $k_z = 9$	$0.524 > 1/2$; tri $= 7.71$	kills $\frac{1}{2}$ off MAIN
χ_3 15	$1.021 > M$; tri $= 3.79$	formula shows death
χ_4 20	$0.892 > M$; tri $= 4.62$	formula shows death
false $F = 2/5$ at $k = 4$	$0.354 \ne 0.157$	head zone is the $F = 1/5$ mask
drop-chain at $k = 4$	$\|t_{\text{loc}}\| = 0.037$	chain is load-bearing
$\psi \times 2$	$2 \cdot \text{tri} \ge 4.48$	majorant worse
false anchor $k_z = 16$	0.756; tri $= 3.07$	formula reproduces the 42-rung razor; the bound is selected-only

6 What this does not do

This does not prove cofinal $|Z_{\text{loc}}| \le \frac{1}{2}$, does not construct k_0, and does not move the mincut. The unsigned, Chebyshev, and mild-Abel routes are named dead, including on selected. No L^* claim, no $R^\dagger$ claim, no RH claim.

References

[1] Round 378, `compose_lemma`.

[2] Round 383, `compose_premises`.

[3] Round 386, `compose_premises2`.

[4] Round 421, `reserve_limit`.

[5] Round 428, `qn_reopened`.